Kavyanjali Agnihotri

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Research Interests

NLP and AI Safety: LLM robustness, representation engineering, agent security, and efficient LLMs. Also NLP for medicine, focusing on reliable AI deployments.

Education

Eberhard Karls University of Tübingen **Tübingen, Germany**
M.Sc. Machine Learning | German Grade = 2.26 *Oct 2024 – Mar 2027 (expected)*

- **Relevant Coursework:** AI Safety, LLM Inference Optimization, NLP, Probabilistic ML, Reinforcement Learning.

Birla Institute of Technology and Science (BITS) Pilani **Hyderabad, India**
B.E. Computer Science | CGPA: 8.49 / 10 ≈ German Grade 1.8 *Nov 2020 – Jul 2024*

Research Projects

Automated Defence Discovery against Prompt Injection | *M.Sc. Thesis* **Jul 2026 – Present**

- *Question:* Defenses tested on static attacks artificially inflate security (e.g., Meta-SecAlign-8B reports ∼0% success but falls to gradients). Investigating if LLM agents can discover defenses that survive white-box re-optimization.
- *Pilot:* a 4-bit Qwen3.6-35B agent tuning Llama Guard 3 (1B) raised its safety-benchmark score (0.8708→0.8882) by cutting false alarms on harmless prompts, while recall on harmful XSTest prompts fell from 75% to 71%.
- *Attacks:* Reproduced 14 *Claudini* gradient attacks against Meta-SecAlign-8B (best breaks 35/50 prompts). On 100 realistic goals, success drops from 21/25 on simple "HACKED" outputs to just 2–3/25 for phishing links or tool calls.
- *Defenses:* Re-implemented sandwich, DataSentinel, TaskTracker and DataFilter. DataFilter stopped 48/48 replayed attacks. Building an LLM-driven attacker to break DataFilter (currently 0/100 success; testing stronger models).
- *Model Tuning:* Reproduced Meta's unpublished SecAlign DPO recipe on Qwen3-4B (torchtune, 4×A100s). Dropped AlpacaFarm attack success from 100% to 8.65% with stable MMLU, yet gradient attacks still succeed.

Skill-File Injection in Open-Weight LLMs | AI Safety **Jun 2026**

- AI Safety course project (team of 3). Extended SkillInject (Schmotz et al., 2026) from proprietary agents to nine open-weight models (Qwen3.5/3.6, Gemma 4) via the OpenHands and vLLM.
- Judge audit: found and fixed silent "safe" verdicts from the LLM judge (a parser bug plus exhausted API quota), raising measured attack success by 24 pp on average and 40 pp at worst. Also found that sandbox containment hid executed attacks from the judge, which led us to add a deterministic script-execution scorer.
- Attack success ranged 3.3–46.4% with no monotonic trend in model size; 34 of 36 injections succeeded on at least one model, and the host skill strongly predicted success (`discord` 63.3% vs. `pptx` 0.8%).

Compositional Understanding in Stable Diffusion | *Vision Language Models* **Sep 2025 – Mar 2026**

- SD3.5 Large Turbo failed on ∼60% of prompts with relational prepositions ("above", "to the right of") in a controlled Self-Bench suite. Built a VLM evaluator (Qwen3-VL-30B, chain-of-thought) that raised agreement with human judgements from 56% (LLaVA) to ∼73%.
- Swapped SD3.5's CLIP-L encoder for compositionally fine-tuned CLIC models without retraining, proving spatial reasoning can be recovered; boosted vertical relation accuracy (e.g., "above") from 72.7% → 90%.

Automatic Generation of Scientific Lecture Notes *(undergraduate research project)* **Jan 2023 – Dec 2023**

- Built a pipeline that turns Wikipedia content (definitions, LaTeX formulae, code) into structured lecture notes, using a knowledge graph of concept dependencies to order topics so prerequisites come first.
- Placed figures and formulae next to the matching text using BERT embeddings, and segmented content into topics with LDA.

Publications

Scholar Inbox: Personalized Paper Recommendations for Scientists **2025**

- M. Flicke, . . . , **K. Agnihotri** (11th of 13), . . . , A. Geiger. *arXiv:2504.08385*, 2025.

Experience

University of Tübingen | Scholar Inbox (Prof. Andreas Geiger) **Tübingen, Germany**
Student Research Assistant (HiWi) *Dec 2024 – Present*

- Benchmarked Qwen3.6 quantized models in llama.cpp vs. vLLM for QA over research papers to measure latency-quality trade-offs within memory limits of an RTX 4090.
- Built LLM tooling for the production pipelines: an admin dashboard to manage, version and monitor prompts.
- Built a pipeline that turns research papers into single-speaker podcast transcripts using Gradium-based audio synthesis.
- Contributed to the full-stack platform (React frontend, Python backend): a frontend design and code update, share buttons, photo upload, and a feature-toggle refactor that stores user preferences in one bitmap column, removing an auxiliary table and simplifying joins. Exploring multi-interest clustering for the digest and scholar-maps views.

Cisco **Bengaluru, India**
Technical Intern *Jan – Jun 2024 & Jun – Jul 2023*

- Fine-tuned a BERT-based question-answering model on internal documentation to answer product-support queries about module updates.
- Built a real-time diagnostic overlay for Secure Network Analytics that surfaces runtime errors from parsed logs.
- Containerised 8+ repositories (Docker, Jenkins CI/CD); upgraded 15+ legacy Java libraries across 20+ microservices to fix dependency vulnerabilities.

Palmtree Infotech | ***Focus: NLP in Medicine*** **Chennai, India**
Technical Intern (NLP) *May – Jul 2022*

- Evaluated GPT-J-6B and GPT-NeoX for Medical Named Entity Recognition (NER), designing zero- and few-shot workflows to extract multi-entity clinical data without fine-tuning.
- Cleaned and restructured complex clinical datasets into document-level JSONL files aligned with GPT-NeoX schemas.

Skills

ML & AI Frameworks: PyTorch, Hugging Face Transformers, vLLM, llama.cpp, scikit-learn

Programming & Web: Python, C++, SQL, Bash, JavaScript (React, Node.js), HTML/CSS

Tools & Infrastructure: Git, Docker, Jenkins CI/CD, MongoDB

Languages: English (Fluent/C2), Hindi (Native), German (A2/B1)

Awards & Achievements

ELIZA Scholarship (2024-2026): Awarded 24-month stipend for ML master studies.

Summer School: Selected for **MLSS 2026** (Reliability & Safety, Kraków).

AI Incubator (Batch 6, 2025): Selected for intensive beginner startup program.

JEE Advanced 2020: All India Rank 4,860 out of 150,000+ candidates (Top 3.2%).